

Statistical Modeling & Evaluation



Intro to Statistical Modeling

Statistical Model is a mathematical model based on a set of assumptions about the probability distribution that generated some observed data

A statistical model is usually specified as a mathematical relationship between one or more random variables and other non-random variables.

Reasons to Learn Statistical Modeling

- It assists in gathering necessary data, doing the proper analysis, and properly presenting the results with statistical understanding.
- It facilitates making scientific discoveries, data-driven judgments, and forecasts.

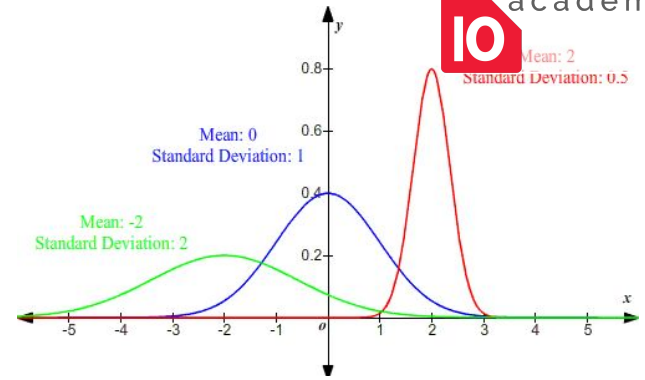
Example Suppose that we randomly draw n individuals from a certain population and measure their height. The measurements can be regarded as **realizations** of n **random variables** $X_1, \dots, X_n$. In principle, these random variables could have any probability distribution. If we assume that they have a **normal distribution**, as it is often done for height measurements, then we are formulating a statistical model: we are placing a restriction on the set of probability distributions that could have generated the data.

Example Suppose that for the same n individuals we also collect weight measurements $Y_1, \dots, Y_n$, and we assume that there is a linear relation between weight and height, described by a **regression equation**

$$Y_i = \alpha + \beta X_i + \varepsilon_i$$

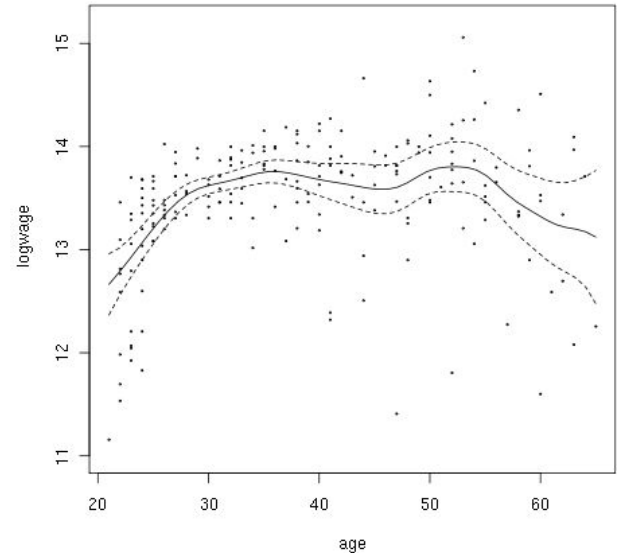
where α and β are regression coefficients and ε_i is an error term. This is a statistical model because we have placed a restriction on the set of joint distributions that could have generated the couples (Y_i, X_i) : we have ruled out all the joint distributions in which the two variables have a relation that cannot be described by the regression equation.

Types of Statistical Models



Parametric Vs. Nonparametric

- Parametric: a family of probability distributions that has a finite number of parameters.
- Nonparametric: models in which the number and nature of the parameters are flexible and not fixed in advance.



Cont.

Conduction Vs. Unconditional

- **Conditional Models** (discriminative): the sample is partitioned into **input** and **output** data. The model is obtained by placing some restrictions on the conditional probability distribution of the outputs given the inputs.
- **Unconditional Models** (generative): analyze the joint distribution of inputs and outputs.

Discriminative

Conditional
model

$$P(y|X)$$

Conditional
distribution

Generative

Unconditional
model

$$P(y, X)$$

Joint distribution



Conditional Models

- **Regression model:** A conditional model in which the output variable is continuous. Common regression models include logistic, polynomial, and linear regression models. Use cases include forecasting, time series modeling, and discovering the causal effect relationship between variables.
- **Classification model:** A conditional model in which the output variable is discrete (or categorical). Common models include decision trees, Naive Bayes, nearest neighbor, random forests, and neural networking models.



Predictive and ML models

Conditional statistical models can be used to make predictions of **unseen** outputs given observed inputs.

Predictive Models: Models that also allow us to make such predictions, but without specifying a set of conditional probability distributions. *They are not really statistical models.*

Several models used in the machine learning field belong to the class of predictive models:

1. Decision Trees; A simple model that splits data into branches to make predictions
2. Random Forests: An ensemble technique that builds multiple decision trees and averages the results
3. Gradient Boosting Machines (GBMs); Boosting is a method of converting weak learners into strong learners. In boosting, each new tree is a fit on a modified version of the original data set.
4. Neural networks;



ML vs. Statistical Modeling

Machine Learning Algorithm	Statistical Modeling
Algorithm that can learn from data without the need of rules-based programming	The formalisation of connections between variables in the form of mathematical equations.
Designed to make accurate predictions without the use of explicit programming	Meant to discover and explain the correlations between variables,
ML models can make better predictions, but they are more difficult to comprehend and explain	Some statistical models may generate predictions, but their accuracy is typically subpar because they cannot capture complicated data interactions



Steps to Build Statistical Model

Step 1: Select the statistical model that best meets your needs.

- Decide if you want to handle a specific inquiry or predict based on a large number of factors.
- Consider how many variables do you need to include in the model? What is the relationship between dependent variables and explanatory variables.

Step 2: Data preparation

- Handling Missing Data
- Feature Engineering: More relevant features lead to better model performance. Create new variables (e.g., claim ratio, average claim frequency).
- Encoding Categorical Data: Use One-Hot Encoding or Label Encoding to convert categories into numerical values.



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- Train-Test Split: This ensures that the model is tested on unseen data. Typical split: 70% training, 30% testing (or 80:20)

Step 3: Collect, organize, analyze, interpret and design data using statistical modeling software.



Evaluation

Metrics for Classification

- Accuracy: defined as the percentage of correct predictions for the test data.
- Precision: defined as the fraction of relevant examples (true positives) among all of the examples which were predicted to belong in a certain class.
- Recall: defined as the fraction of examples which were predicted to belong to a class with respect to all of the examples that truly belong in the class.
- The F-score or F-measure is a measure of predictive performance. It is calculated from the precision and recall of the test.



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Evaluation metrics for Regression include

- Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE)



Interpretation

- ML model can act as a “Black Box”: arrive at correct predictions, but we don’t understand how they arrived at those predictions.
- There are tools that can help
 - SHAP (SHapley Additive exPlanations): values are a way to explain the output of any machine learning model. It uses a game theoretic approach that measures each player's contribution to the final outcome. In machine learning, each feature is assigned an importance value representing its contribution to the model's output
 - LIME (Local Interpretable Model-agnostic Explanations): LIME focuses on explaining the model's prediction for individual instances



REFERENCES

- <https://www.statlect.com/glossary/statistical-model>
- <https://www.simplilearn.com/tutorials/statistics-tutorial/what-is-statistical-modeling>
- <https://www.coursera.org/articles/statistical-modeling>
- <https://www.datacamp.com/tutorial/explainable-ai-understanding-and-trusting-machine-learning-models>
- <https://shap.readthedocs.io/en/latest/index.html>